

4.7. ch5 刘亦优 12613149 朱金龙

7. (a) $x = -|\vec{F}_1| - m|\vec{a}| \sin \theta = -32.0 \text{ N}$
 $y = -m|\vec{a}| \cos \theta = -20.8 \text{ N}$
 so $\vec{F}_2 = (-32.0 \hat{i} - 20.8 \hat{j}) \text{ N}$
 (b) $|\vec{F}_2| = \sqrt{32.0^2 + (-20.8)^2} = 38.2 \text{ N}$
 (c) suppose the angle is ϕ
 so $\tan \phi = \frac{20.8 \text{ N}}{32.0 \text{ N}} = 0.65$
 so $\phi = 213^\circ$

28. $T = 1.91 \times 10^4 \text{ N} > m_B g = 1.176 \times 10^4 \text{ N}$, so go up.
 $m_B a = T - m_B g \Rightarrow a \approx 6.12 \text{ m/s}^2$
 $ma = N - mg \Rightarrow N \approx 191 \text{ N}$

58.
 (a) $2T_1 = mg \Rightarrow T_1 = 505 \text{ N}$
 (b) $2T_2 = mg = ma \Rightarrow T_2 = 572 \text{ N}$
 (c) $T_3 = mg = 1010 \text{ N}$
 (d) $T_4 - mg = ma \Rightarrow T_4 = 1140 \text{ N}$
 (e) $F_1 = 2T_1 = 1010 \text{ N}$
 (f) $F_2 = 2T_2 = 1140 \text{ N}$
 (g) $F_3 = 2T_3 = 2020 \text{ N}$
 (h) $F_4 = 2T_4 = 2280 \text{ N}$

13. For every moment: $x_1 + x_2 = 2x_3 \Rightarrow v_1 + v_2 = 2v_3 \Rightarrow a_1 + a_2 = 2a_3$

$$\begin{cases} ma_1 = T \\ 2ma_2 = T \\ 3ma_3 = 3mg - 2T \\ a_1 + a_2 = 2a_3 \end{cases} \Rightarrow \begin{cases} a_1 = 6.92 \text{ m/s}^2 \\ a_2 = 3.46 \text{ m/s}^2 \\ a_3 = 5.19 \text{ m/s}^2 \end{cases}$$

55. $F = (m_1 + m_2)a \Rightarrow a \approx 0.91 \text{ m/s}^2$
 (a) $F_1 = m_2 a = 1.1 \text{ N}$
 (b) $F_2 = m_1 a = 2.1 \text{ N}$
 (c) The masses are different but the acceleration doesn't change, so the values are different.

67.
$$\begin{array}{c} \uparrow a \\ T_1 = m_A g \quad T_2 = m_C g \\ \leftarrow \quad \rightarrow \end{array} \quad \begin{aligned} m_C g - m_A g &= (m_A + m_B + m_C)a \\ \Rightarrow a &= 1.63 \text{ m/s}^2 \\ m_C g - T_2 &= m_C a \Rightarrow T_2 &= 81.7 \text{ N} \end{aligned}$$